NumPy & Pandas - One Page Quick Notes

NumPy
- Library for numerical computing in Python.
- Main object: ndarray (N-dimensional array).
- Create arrays: array(), zeros(), ones(), arange(), linspace().
- Useful operations: reshape(), flatten(), mean(), sum(), min(), max(), dot(), transpose().
- Supports vectorized operations for fast computation.

Pandas
- Library for data manipulation and analysis.
- Main objects: Series (1D) and DataFrame (2D).
- Read data: read_csv(), read_excel().
- View data: head(), tail(), info(), describe().
- Select data: loc[], iloc[].
- Handle missing values: isnull(), fillna(), dropna().
- Data operations: groupby(), merge(), sort_values(), value_counts().
- Save data: to_csv(), to_excel().

Difference
NumPy focuses on numerical arrays and mathematical operations, while Pandas provides labeled tables and powerful data analysis tools built on top of NumPy.